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Identifying Symmetries in Quantum Machine Learning with LLM-Guided Evolutionary Algorithms

Machine learning is a widely used technique to discover patterns in a dataset. Classical machine learning architectures contain trainable weights, whose values can be modified so that the model could learn the features of a dataset. Quantum computers, on the other hand, have become a promising emerging technology. Unlike classical computers, whose bits are either 0 or 1, quantum bits, or qubits, are in states which are superpositions of $|0\rangle$ and $|1\rangle$. This allows more information to be represented with the same amount of resources. When it comes to machine learning, this gives quantum computers a significant advantage: to represent the same number of features in machine learning, around 50 quantum bits can surpass the best supercomputers we have today. Quantum machine learning (QML) uses quantum circuits, which composes of quantum gates that modifies qubit states. The parameters, often rotation angles around some specified axis, can thus be tuned so that the circuit learns a pattern from the training dataset.

Similar to natural evolution, evolutionary algorithms are methods to find solutions of a problem whose search space is highly dimensional. In this work, we use ShinkaEvolve, an open source framework for LLM evolutionary search. Given a quantum machine learning problem, an ensemble of three LLMs continuously generate quantum circuit designs, and the objective is to optimize the score of the circuit, generated by an independent evaluator script. The ensemble chooses one LLM to propose a quantum circuit. Then the model can choose to make a small modification from the previous circuit, combine two programs to make a new one, or propose the scratch and propose its own circuit.

I started this research by trying to find a suitable quantum machine learning problem to run this evolutionary algorithm on. The first problem I discovered is a tic-tac-toe problem: given the state at each of the nine cells in a standard tic-tac-toe board, which side is winning, or is it a draw? After some deeper investigation, I realized that the significance of this particular problem is that there are a lot of built in symmetries. For example, rotating the board by 90 degrees does not change the solution at all. Therefore, the optimal quantum circuit that solves this problem should be composed of several layers that satisfy the symmetry requirements of the problem. Inspired by this, I found a QML problem with a simpler symmetry. Consider a graph of 8 nodes, where each of them can be connected or disconnected. Given the connectedness of the 28 pairings, the problem is to find whether one can reach any of the eight nodes from any of the other nodes. The problem is symmetric because shuffling the orderings of the eight nodes should not change the solution. This QML problem thus became the focus of my research.

I used three sets of LLMs, each with different capabilities, in the evolutionary search, and compared their results when asked to find an optimal solution to this problem. I discovered that the strongest ensemble is able to identify the optimal configuration in the first few generations, while the other ensembles struggle to find it and show inconsistent results across multiple runs. In addition, the frontier models always build what they claim to have built in the descriptions. Meanwhile, the mid-tier ensemble only does this around half the time, and the descriptions of the weakest models rarely matches what they have built. This stark difference can be attributed to the structure of the evolutionary algorithm. Mistakes will compound over time, but correct progress will lead to higher and higher scores.

This work demonstrates that while LLMs can be useful in identifying symmetries to a QML problem, one must take caution in using them. A researcher should run the same set of models several times and only claim that a symmetry exists if such a symmetry exists in all of the runs. If the LLMs claim the existence of a symmetry only in a few runs, then that is likely to be an accidental setup, not something inherent to the problem. Future work can aim to quantify the capabilities of LLMs in evolutionary search on different problems, ranging from mathematical proofs, drug discovery, machine learning optimization, to competitive programming. I plan to write up the results of this work formally and submit a preprint to Arxiv.

I thank all the donors and supporters of the Yale College First-Year Summer Research Fellowship in the Sciences & Engineering. Every number in my work requires calling LLMs through APIs, which costs anything ranging from \$1 to \$10 per run. The project also went through a lot of trial and error, and I would not be able to bear the cost of generating results on my own were it not for this fellowship. In addition, I also participated in all of the workshops where I collaborated with fellow students, read a paper, and explained it with a 40-minute presentation. This enhanced my ability in understanding novel concepts quickly, which will be beneficial in my future research endeavors.